

Where should the one new line go?

optimal single-edge placement in a second-order Kuramoto network: a cheap frequency-domain cost, and the risk it does and does not predict

Abstract

Given a fixed grid, a fixed operating point and permission to build exactly one new circuit of strength κ , which pair of buses should it join? We compute a *cost* C — an $\mathcal{H}_\infty$ peak of the disturbance-to-frequency gain, cheap enough to evaluate for all 4950 candidate pairs — and then ask whether it predicts *risk* R , meaning what the grid actually does, measured as the companion SNSP study measures it. The answer is sharply split. The best edge is a long line from the north-west corner into the largest synchronous machine. Built once and applied to all 240 operating states of that study’s ensemble, it lowers worst-five-bus RoCoF in 207 of them, by 3.9 % on average and 8.6 % at the configuration it was chosen on — the swarm moves left by a median of **5.8 points of SNSP** — while worst-five-bus frequency deviation barely moves (0.6 %, 0.8 points). The cost gets this backwards: it tracks the deviation risk well (ρ up to +0.85) and the RoCoF risk not at all (ρ between -0.22 and $+0.14$). It is well correlated with the risk an edge cannot move, and uncorrelated with the risk it can. The benefit also disappears above 85 % SNSP, where the machine it attaches to is often not running.

1 Two quantities, and why they must not share a letter

- **The cost** C is a property of the Green’s function $G(s) = A(s)^{-1}$: the peak over a frequency band of $\sigma_{\max}$ of the disturbance-to-frequency gain. It costs about 0.5s to evaluate for all 4950 candidates at one frequency, which is what makes an exhaustive search possible. It is *not* a risk. It is a quantity we reason physically *ought* to correlate with risk, and §5 is where that reasoning is tested rather than assumed. Three variants: C_{freq} (frequency-deviation weighting), C_{rocof} (windowed-RoCoF weighting) and C_{H_2} (the band-limited $\mathcal{H}_2$ norm).
- **The risk** R is what the grid actually does. Integrate the full nonlinear-operating-point response to the frozen disturbance set and measure the 500 ms RoCoF and the peak frequency excursion *at the five worst buses*, averaged over events (R_{rocof5} , R_{dev5}). These are the vertical axes of the SNSP study’s own `fig11_topbus_vs_predictor`, i.e. the quantities that study drew its conclusions on, and the ones an added edge must be judged against. Single-worst-bus versions are carried alongside, because a grid code binds on one bus.

2 The model and the decision

The state is the linearised swing equation with a governor at every committed machine. Eliminating the governor state p from $M\ddot{x} + D\dot{x} + Lx = \Delta P + p$, $T\dot{p} = -p - \text{droop } \dot{x}/\omega_s$ gives

$$A(s) = Ms^2 + \left[D + \frac{\text{droop}}{\omega_s(1+Ts)} \right] s + L, \quad G(s) = A(s)^{-1}, \quad (1)$$

i.e. the spec’s $Ms^2 + \Gamma s + L$ with a *frequency-dependent* diagonal damping. $A(s)$ remains complex symmetric and diagonal-plus-Laplacian, so the rank-one machinery is unaffected. L carries synchronising coefficients $L_{ij} = K_{ij} \cos(\theta_i^* - \theta_j^*)$, not bare couplings.

The grid is 10×10 , 166 lines, wind on the Atlantic seaboard (four farms radial), demand east and south, five synchronous stations. The operating point is configuration #128 of the SNSP ensemble: 70 % SNSP, two of five machines synchronised, 33.79 p.u. s stored energy. Adding edge (a, b) is the rank-one update $A \rightarrow A + \kappa uu^\top$, $u = e_a - e_b$, with $\kappa = 2.5$ (one new spur).

3 Why there is a band at all, and why it decides the answer

This is the part of the procedure that is easiest to mistake for a tuning knob, so it is worth setting out in full.

The band is part of the definition of the cost, not a parameter of it. An $\mathcal{H}_\infty$ norm *is* a maximum over frequency. Writing $C = \max_\omega |\Omega(\omega)| \sigma_{\max}(\Pi G(i\omega)B)$ is meaningless until one says which ω are admitted. The reason to take a maximum at all is that we do not know the spectral content of the disturbance: the worst-case response to a bounded input of unknown shape is the largest gain the system has anywhere the input might live. (If one *did* know the spectrum, the right object would be an integral against it — that is $\mathcal{H}_2$, computed here too.) So a band is unavoidable, and it encodes a modelling claim: *these are the frequencies at which a disturbance might have power and at which the model is telling the truth.*

It decides the answer because the gain is not flat. $\sigma_{\max}(\omega)$ is a resonance spectrum (Figure 1). A maximum over such a function is, to good approximation, *the height of the tallest resonance admitted*. Moving the band edge across a resonance therefore does not perturb the cost; it re-points it at a different mode — and a different mode means a different physical mechanism, hence a different best edge. The band is the switch that selects which oscillation the optimiser is asked to suppress.

On this grid the switch sits between physics and numerics. The SNSP model gives an algebraic bus a small regularising mass so the equations stay an ODE. Ninety-three of the 100 buses carry $M \approx 6 \times 10^{-4}$, and they ring against their own circuits at $\sqrt{K/M}$, populating everything above 2.5 Hz with modes that a real converter bus does not have. A time-domain integral barely weights them — the SNSP study ran these same buses throughout without difficulty — but an $\mathcal{H}_\infty$ peak seeks out the largest gain in the band and lands on one. The diagnostic is direct: a regularisation mode scales as $1/\sqrt{M}$ and runs away as the regularising mass is reduced, while a physical mode converges.

mode at H	at $H/2$	at $H/4$	verdict
0.086 Hz	0.091	0.093	physical
1.827 Hz	1.884	1.909	physical (converging)
2.543 Hz	3.365	4.574	regularisation ($\approx 1/\sqrt{M}$)
4.10 Hz	5.771	—	regularisation

Table 1: Halving the regularising mass on the light buses separates the two families cleanly. The band is cut automatically at the geometric mean of the highest physical and lowest regularisation mode, **0.01**–2.155 Hz, which keeps the former’s resonance peak inside and leaves the latter’s outside.

The consequence is quantitative and large. Scored to 5 or 20 Hz, the no-edge peak moves from 1.831 to 2.556 Hz and the ranking is not perturbed but *unrelated*: Spearman 0.337 on C_{freq} , -0.063 on C_{rocof} , 0/10 overlap in the top ten, and a different winner (Table 2). The 5 and 20 Hz rows are identical to each other, because both admit the same tallest artefact.

4 Procedure

1. Freeze everything except the edge.

- Load the grid with a SHA-256 check, so a quietly edited network cannot contaminate a comparison.

- Take P , H , droop and T_{gov} from one ensemble configuration and never move them. The steady state of the swing equation depends on P and K but never on M , so at fixed P the added edge is the only variable.
 - Enumerate all $\binom{100}{2} = 4950$ pairs: 4784 new, 166 reinforcements of existing circuits kept as a comparison class.
- 2. Verify before believing** — 24 checks, including all nine of the design spec’s §10:
- Sherman–Morrison against a direct inverse (worst relative error 2.7×10^{-12}); the Crabtree–Haynsworth quotient identity (4.4×10^{-15}); the Guyan limit; the norm ordering $\max_{ij} |G_{ij}| \leq \sigma_{\max} \leq \|G\|_F$.
 - Frequency domain against a time-domain simulation of the same system: agreement to six significant figures.
 - Both structural degeneracies, asserted *with an edge added*: the settling offset $\sum \Delta P / \sum (D + \text{droop}/\omega_s)$ and the initial RoCoF $\Delta P_j / m_j$ are unchanged to the digit.
 - That the base case *reproduces the corresponding row of the SNSP study’s own results*, exactly. This is not decoration: the frozen `configs.npz` in that folder is stale, carrying a droop vector summing to 40 where the study’s own redraw gives 114. Reading it yields a differently damped grid whose RoCoF still matches to 0.5% but whose $\int f^2$ is 2.5× too large, and droop enters (1) directly.
- 3. Choose the band** by the test of §3.
- 4. Define the cost.** With $\Pi = I - \mathbf{1}\mathbf{1}^\top M / (\mathbf{1}^\top M \mathbf{1})$ the centre-of-inertia projector and B the 20 frozen SNSP disturbances,

$$C_{\text{freq}} = \max_{\omega \in [\omega_1, \omega_2]} |\Omega_f| \sigma_{\max}(\Pi G_{\text{new}}(i\omega) B), \quad \Omega_f = \frac{s}{2\pi(1 + sT_{\text{meas}})}, \quad (2)$$

and C_{rocof} the same with the extra windowed-RoCoF gain $2|\sin(\omega T_w/2)|/T_w$, with $T_{\text{meas}} = 100$ ms, $T_w = 500$ ms.

- Both weights are scalars times the identity, so they factor out of the singular value: *one* $\sigma_{\max}(\omega)$ sweep serves both, and what separates them is only *where in the band* each peaks.
 - RoCoF is windowed, never instantaneous: a window of length T has gain $2|\sin(\omega T/2)|/T$, not ω .
- 5. Solve once, store the curves.** Per frequency: one $O(N^3)$ inverse, then per candidate a rank-one update of the 100×20 matrix ΠGB and the top eigenvalue of its 20×20 Gram matrix (≈ 0.5 s for all 4950).
- No shortlist: every candidate can afford the real cost, so the $\mathcal{H}_2$ stage is run to *test* the spec’s claim that it is a screen and not a criterion, rather than to filter.
 - $\sigma_{\max}$, $\|\cdot\|_F$ and $\max_i \sum_j |\cdot|$ are stored for every candidate at all 426 frequencies, so band, measurement lag, RoCoF window and norm all become *reprocessing*.
- 6. Test the ranking against the scoring choices** — band, lag, window, grid resolution — by re-scoring the stored curves.
- 7. Re-solve a shortlist honestly, and measure the risk.** For 45 candidates (top of each ranking, the worst by C_{freq} , and a random sample):
- Rebuild the grid with the edge present, re-solve the load flow, rebuild L from the new θ^* , recompute (2) directly. This matters: one edge changes **422 entries of L** , not the four the rank-one form touches, and moves θ^* by up to 0.56 rad.
 - Integrate the response to the same 20 events and reduce it to R_{rocof5} , R_{dev5} and their single-bus counterparts.
- 8. Build the winning edge and re-run the entire ensemble through it.** One chosen edge, all 240 balanced operating states, same frozen disturbances — so the benefit is a shift

of a population rather than one arrow on one point, and so that an edge chosen at 70 % SNSP can be shown not to lose at 40 %. The unmodified swarm is the published SNSP result, re-verified against this folder’s code path before use.

5 Results

5.1 The response, and what the band admits

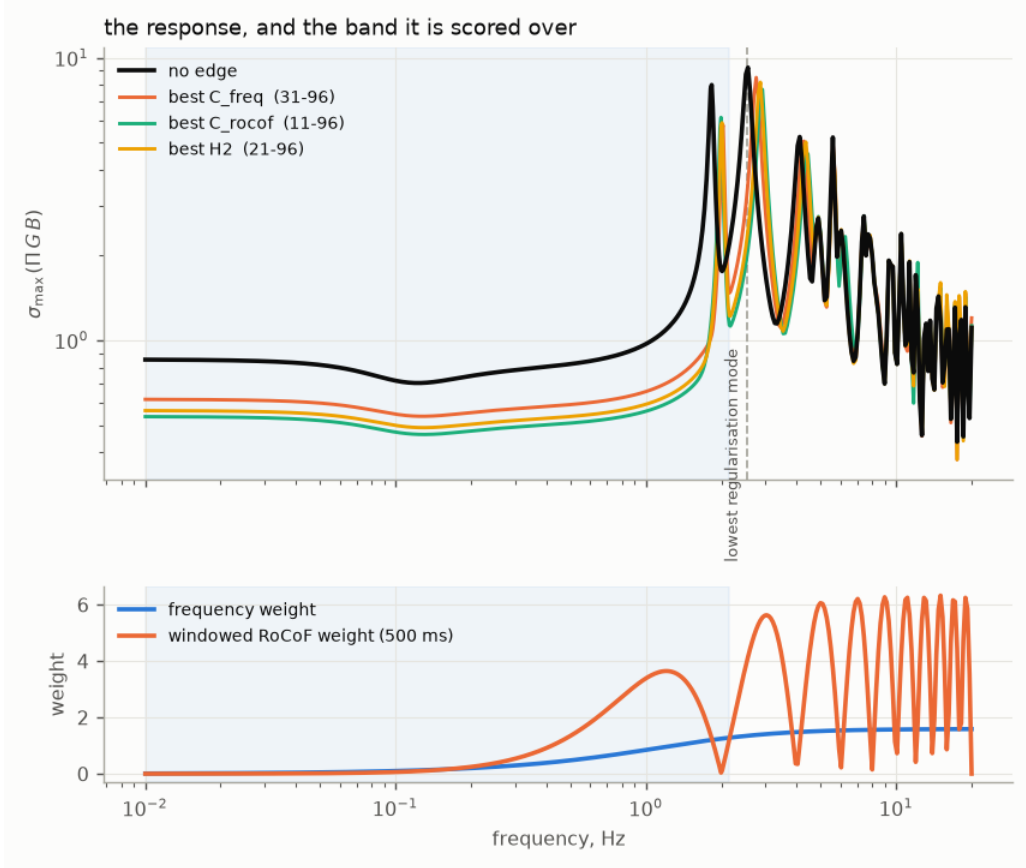


Figure 1: $\sigma_{\max}(\Pi GB)$ for the base network (black) and the three winners, with the scored band shaded and the first regularisation mode dashed; below, the two measurement weights. (i) Below 1 Hz every candidate lowers a broad plateau from 0.75 to ≈ 0.45 : the genuine, broadband part of what an edge buys. (ii) The forest of resonances above 2.5 Hz is the regularisation, and all candidates lie on top of one another there — nothing an edge does reaches it, which is exactly why admitting it into the band destroys the ranking. (iii) The RoCoF weight (orange, lower panel) has nulls at every multiple of $1/T_w = 2$ Hz and does *not* roll off, so scoring beyond the shaded band means scoring the sidelobes of the instrument against the artefacts of the model.

5.2 Every good candidate is the same edge

Of the 4784 new edges, 34.6 % *raise* C_{freq} (worst $\times 1.087$) and 7.6 % *raise* C_{rocof} : Braess is unambiguously present. Spearman between C_{freq} and C_{rocof} is $+0.701$ with only 4/20 of the top twenty shared, yet the Pareto front (Figure 3) is three points long — the scalarisation is nearly irrelevant among the best candidates and matters greatly in the bulk. Against the $\mathcal{H}_2$ screen, $\rho = +0.891$ with 11/20 shared, so on this grid $\mathcal{H}_2$ would have served; on the artefact band it was $\rho = -0.514$ with 0/20. The screen’s reputation depends on the band, not on the norm.

variant	Spearman vs default	top-10 shared	best available
band 0.01–5 or 20 Hz, C_{freq}	0.337	0/10	0.847
band 0.01–5 or 20 Hz, C_{rocof}	−0.063	0/10	0.988
measurement lag 50 ms	0.989	8/10	0.771
measurement lag 200 ms	0.991	9/10	0.731
RoCoF window 250 ms, C_{rocof}	0.671	1/10	0.757
every second frequency point, C_{freq}	0.993	1/10	0.685

Table 2: Robustness of the *cost* ranking, all by re-scoring stored curves. The measurement lag is harmless, as it is for the SNSP cost. The band is decisive. The last row is a caveat: the overall order is stable but the top ten reshuffles, and a coarser grid finds a *better* best because subsampling misses peaks — the field near the top is not resolved to better than a few per cent.

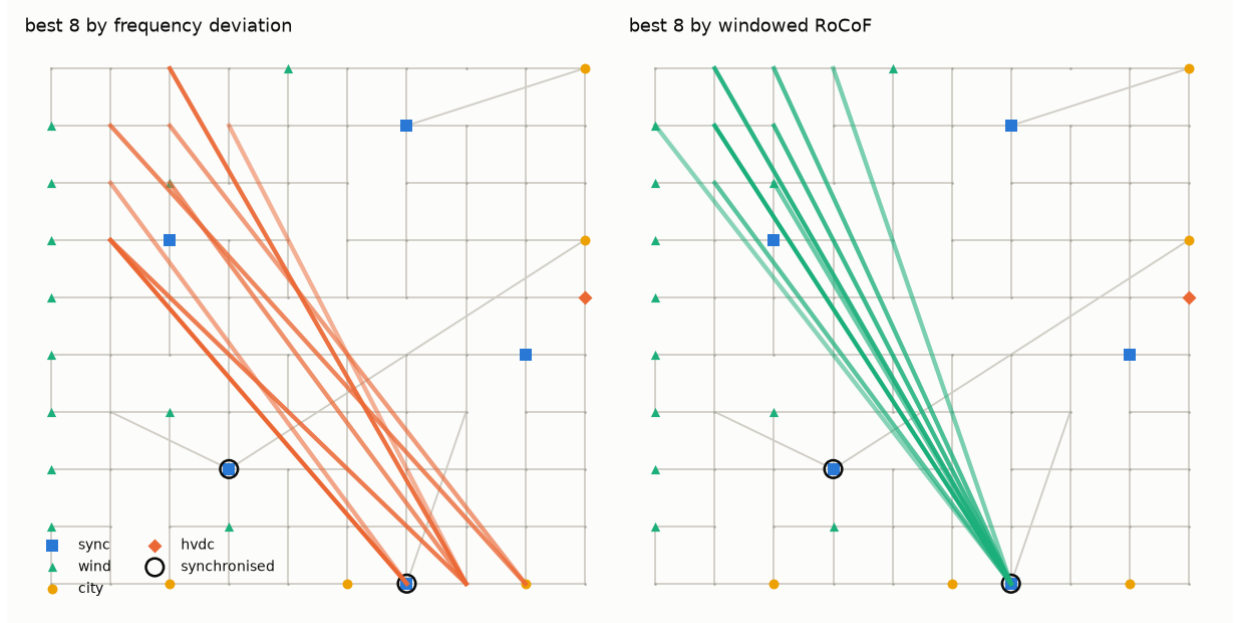


Figure 2: The best eight edges under each cost. Almost all terminate at bus 96 or its neighbour 97 — the synchronous machine at row 9, column 6, carrying $H = 4$ s (ringed markers are the two machines actually synchronised). The other end is always the north-west corner, where the radial wind spurs sit. The $\mathcal{H}_\infty$ peak at 1.827 Hz is the mode in which bus 96 swings against the rest of the system, so the cost’s entire answer is “tie the far corner to the biggest mass”. This configuration has only two physical modes below the regularisation floor — 0.086 and 1.827 Hz — which is why the answer is so monolithic.

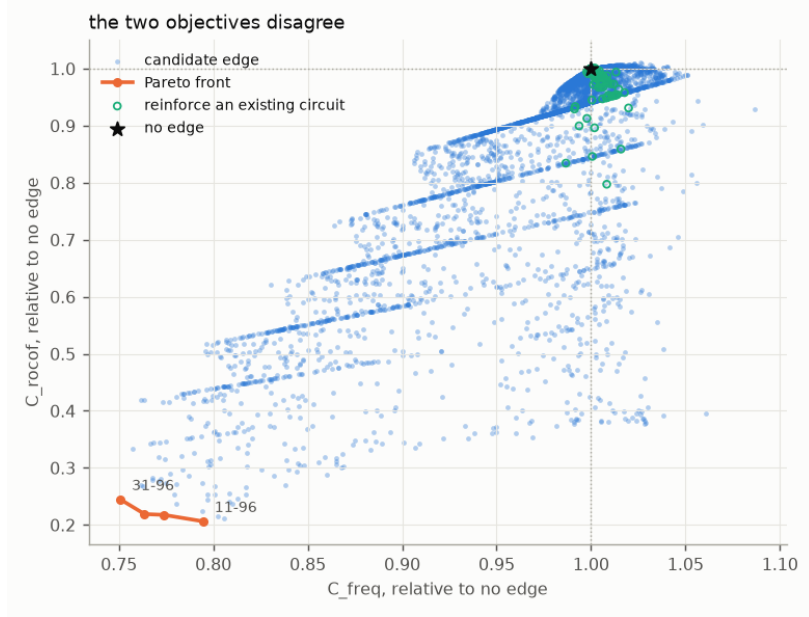


Figure 3: Every new edge in $(C_{\text{freq}}, C_{\text{rocof}})$, relative to no edge; the star is the base network. The striations are candidates sharing an endpoint: the bus one attaches to sets the band, the other end fine-tunes it.

5.3 What the edge actually buys

	mean	median	best	worst	lower in
R_{rocof5}	0.9615	0.9588	0.8709	1.0584	207/240
R_{dev5}	0.9945	0.9943	0.9871	1.0022	220/240
RoCoF, single worst bus	0.9666	0.9655	0.8453	1.0629	175/240
<i>mean R_{rocof5} ratio by SNSP band</i>					
30–50 %	0.9429				
50–70 %	0.9480				
70–85 %	0.9575				
85–100 %	0.9966	(single worst bus: 1.0106)			

Table 3: One edge, chosen at a single operating point, applied to all 240. It survives the sweep — but it stops working at the top. Above 85 % SNSP the benefit vanishes and the single-worst-bus RoCoF turns slightly negative, because bus 96 is often no longer synchronised there, and an edge into a decommitted machine buys nothing. That is precisely the region an operator would want reinforcement to help in.

Configuration #128 is a better-than-average case: the same edge is worth 8.6 % there against 3.9 % averaged over the ensemble. Figure 5 shows the single-configuration view, with the sideways reading done against the mean curve.

The asymmetry has a clean explanation. Frequency deviation is dominated by the common-mode droop offset $\Delta P / \sum (D + \text{droop} / \omega_s)$, which is topology-independent by construction and which no edge can touch; what an edge changes is the differential motion between buses, and that is what a 500 ms RoCoF at the worst buses measures.

5.4 The cost predicts the wrong risk

Two supporting observations. First, the rank-one screen is a decent but not reliable proxy for the exact cost: screen against exact re-solve gives $\rho = +0.786$ on C_{freq} , $+0.933$ on C_{rocof} and $+0.870$ on C_{H_2} . Second, and more tellingly, candidate 46–73 was selected *because the screen judged it harmful* ($C_{\text{freq}} \times 1.051$) and turns out to be the fourth-best edge in the whole shortlist by R_{rocof5}

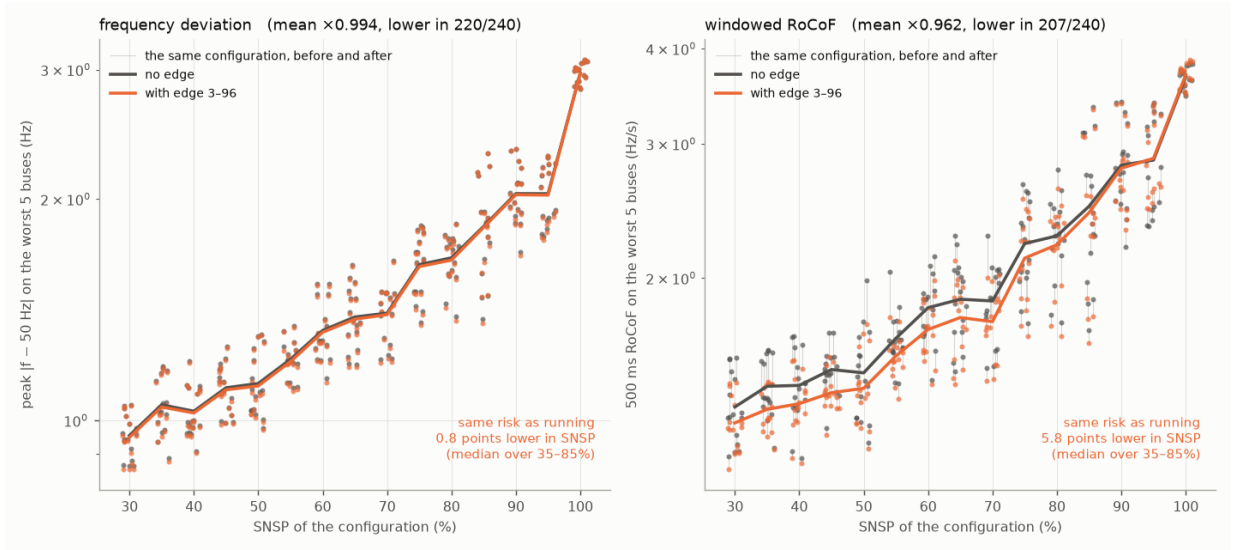


Figure 4: The benefit as a shift of a whole population. Every one of the 240 balanced operating states of the SNSP ensemble is solved twice — once on the frozen grid (grey) and once with edge 3–96 built (orange) — against the same frozen disturbances, with each configuration keeping its own commitment, inertia and wind pattern. **The movement is purely vertical.** SNSP is a property of the dispatch, the dispatch is frozen, and an added edge cannot touch it: every configuration sits at exactly the same share before and after, verified to 0.000×10^0 . Each pair is therefore drawn as a vertical segment, and the scatter is jittered only to separate overlapping points, with the same jitter applied to both swarms. Lines are the mean at each level. The benefit can nonetheless be read *sideways*, as an equivalence rather than a movement: the RoCoF gain is what running **5.8 points lower in SNSP** would have bought, which is a number an operator can price, because points of SNSP are curtailment. For deviation it is 0.8 points. The “before” swarm is the published SNSP result, reproduced by this folder’s code to 2.2×10^{-16} before the comparison was drawn.

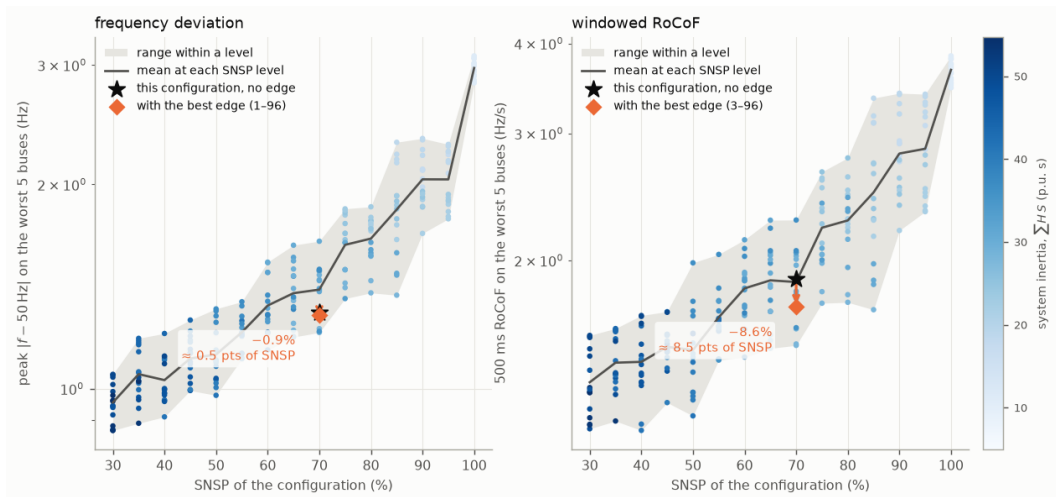


Figure 5: The same comparison for configuration #128 alone, on the layout of the SNSP study’s own `fig11.topbus_vs_predictor`: the ensemble shaded by system inertia, with this configuration before (star) and after (diamond).

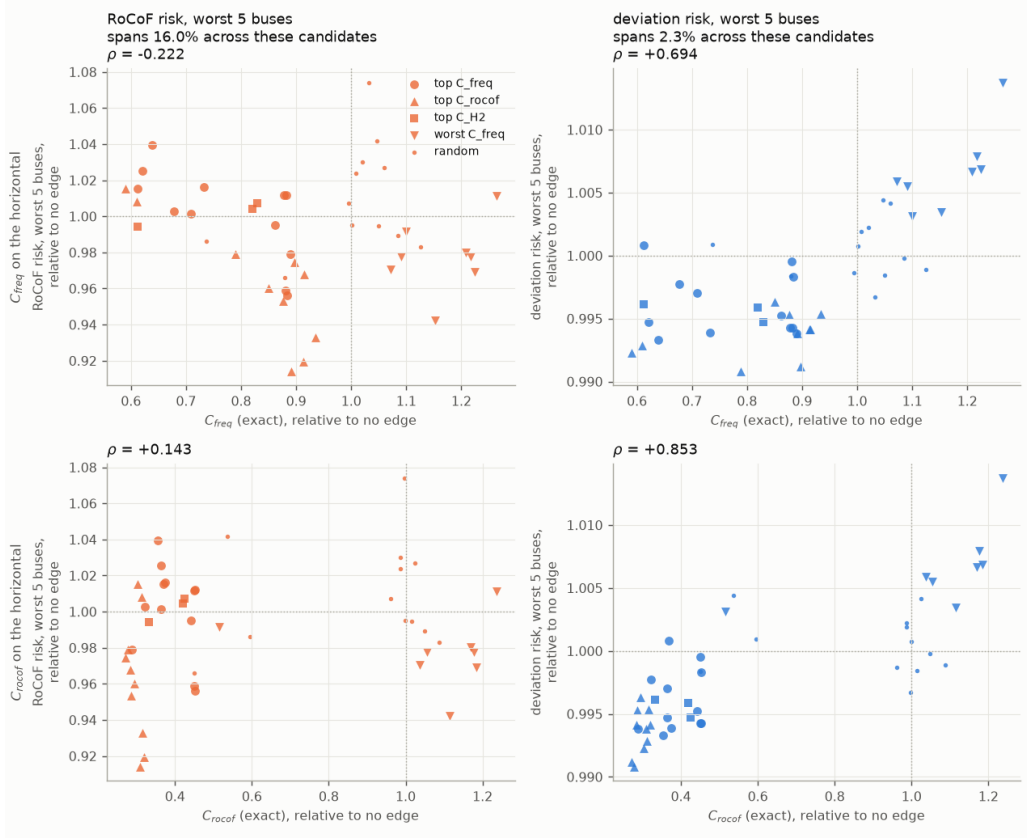


Figure 6: The same 45 candidates. Rows share a cost — C_{freq} on the horizontal above, C_{rocof} below — and columns share a risk. **The two columns do not share a vertical scale, and cannot:** an edge moves RoCoF risk over a range seven times wider (16.0% of the no-edge value, against 2.3% for deviation), so a common axis would flatten the right-hand column into a line. Each column heading gives its span, so the two are compared by number rather than by eye. That difference in span is itself the substantive point, and it is worth stating plainly: it does not mean deviation risk is unimportant, but that on this grid *nothing an edge does changes it much*, because it is set by the topology-independent droop offset. RoCoF risk is the one an edge has real leverage over — and it is the one no cost predicts. Markers record why each candidate was selected. The split of Table 5 is then visible directly — down the left-hand column neither cost has a usable slope against RoCoF risk, while down the right both have a clear one against deviation risk. *The bottom row is the control that matters.* C_{rocof} is the cost that *carries the windowed-RoCoF weight*, so if any of these quantities were going to predict RoCoF risk it would be that one; it manages $\rho = +0.143$, no better than the cost that ignores RoCoF entirely, while scoring its best correlation (+0.853) against the deviation risk it was not built for. The failure is therefore not a matter of choosing the wrong weighting.

risk measure	no edge	best of 45	worst of 45	best edge
R_{rocof5} , 500 ms RoCoF, worst 5 buses	1.8860 Hz s ⁻¹	×0.9142	×1.0741	3–96
R_{dev5} , peak $ f - 50 $, worst 5 buses	1.2959 Hz	×0.9908	×1.0137	1–96
RoCoF, single worst bus	0.4054 Hz s ⁻¹	×0.9178	×1.0958	3–96
peak $ f - 50 $, single worst bus	0.2777 Hz	×0.9923	×1.0093	1–96
nadir, worst 5 buses	1.2093 Hz	×0.9902	×1.0140	1–96
J , the SNSP functional	1271.8 Hz ² s	×0.9595	×0.9975	10–96

Table 4: One edge is worth about 8.6% of RoCoF risk and about 1% of deviation risk. Sum and maximum agree here: $\text{Spearman}(R_{\text{rocof5}}, R_{\text{rocof}_{\text{top1}}}) = +0.967$. Nineteen of the 45 candidates *raise* R_{rocof5} and fifteen raise R_{dev5} , so Braess survives into the risk as well as the cost.

	R_{rocof5}	R_{dev5}	RoCoF top1	dev top1	nadir top5
C_{freq}	−0.222	+0.694	−0.226	+0.657	+0.712
C_{rocof}	+0.143	+0.853	+0.123	+0.768	+0.825
C_{H_2}	−0.179	+0.779	−0.186	+0.735	+0.779

Table 5: Spearman correlation of each frequency-domain *cost* against each time-domain *risk*, over the 45 re-solved candidates. Every cost tracks deviation risk well and RoCoF risk not at all — and RoCoF is the risk an edge can actually move (Table 4). Note also that C_{rocof} , the cost carrying the RoCoF weighting, is no better at predicting RoCoF risk than the others; it is merely the best predictor of *deviation* risk.

(×0.942). A post-fault equilibrium exists for every candidate, the worst line reaching 0.358 of its static limit, so angle stability never binds.

6 Caveats and what is open

- **The costs are diagnostics, not objectives — on this evidence.** They rank 4950 candidates for the price of a few minutes, and they are genuinely informative about frequency deviation. But selecting an edge by C_{freq} would have discarded the best RoCoF candidate found here. If RoCoF is the binding constraint, the shortlist must be scored in the time domain, and the cost used only to keep the shortlist affordable — and it should be a *generous* shortlist, including candidates the cost dislikes.
- **One operating point, one κ , one disturbance model.** An edge is a permanent asset scored here at configuration #128. The configuration sweep matters most: this point has only two physical modes, and one with more machines synchronised would offer more structure for an edge to act on.
- **The winner is not resolved** (Table 2, last row). Read “attach the north-west corner to bus 96”, not “build 31–96”; and note that the best edge by risk (3–96) is not the best by cost (31–96).
- **The disturbances are steps.** Both structural degeneracies — settling offset and initial RoCoF — are properties of a step. A disturbance with power in the 0.5–2 Hz band would put energy exactly where an edge has leverage, and the $\mathcal{H}_\infty$ framing would then be the right one. Re-running with narrowband forcing is the single most informative follow-up.
- **No line ratings, no $N - 1$, no voltage.** A new edge here cannot overload, and system strength — the other half of why reinforcement is bought — is absent from a swing-equation model. Distances are lattice squares and κ is a coupling, not a rating.